

**DEPARTMENT OF COMPUTER & INFORMATION SYSTEMS ENGINEERING
BACHELORS IN COMPUTER SYSTEMS ENGINEERING**

Course Code: CS-116

Course Title: Object Oriented Programming

Complex Engineering Problem

FE Batch 2025, Spring Semester 2026

TERM PROJECT

Course Learning Outcome

CLO 2: Apply acquired concepts to analyze, develop and debug programs for real world applications (Taxonomy level C3).

CLO 3: Produce an object oriented design using project management tools. (Taxonomy level C3).

Complex problem-solving attributes (CPA) covered (as per PEC - OBA manual – 2019)

- **CPA-1 Depth of analysis required:** Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.
- **CPA-2 Level of interaction:** Require resolution of significant problems arising from interactions between wide ranging or conflicting technical, engineering or other issues.
- **CPA-3 Familiarity:** Can extend beyond previous experiences by applying principles-based approaches.

Problem Statement

Develop an object-oriented programming application in C++.

Design Constraints

- Organize the application into at least **4** interlinked classes.
- Your application must exhibit at least **5** of the following object-oriented features:
 - Inheritance
 - Association (Aggregation/Composition)
 - Method overriding
 - Operator overloading
 - Abstract classes
 - Exception handling
- Organize your code in classes, functions/methods and files.
- The code must be easy to read and follow. Provide concise and useful comments generously.
- You may use any built-in library functions, external library/package or framework as long as the resulting code is meeting the given requirements as specified above.

Instructions and Guidelines (Read carefully)

1. **Students can work in groups of 4 at maximum and 3 at minimum. All members must belong to the same practical group. Group work and co-ordination carry marks, so individual work is discouraged.**
2. The project details are included in this document. You can add any additional features to the project to make it distinct and stand out.
3. Submit your file(s) in a zip folder with the following naming convention: GX-YY_CEP, where X represents your practical group and YY represent your group number (as mentioned in GCR). For example: G3-12_CEP.zip
4. Since the folder would contain more than 1 file (main code file + transaction data file etc.), name your files accordingly.
5. Add a ReadMe file that assists with project setup, as the teacher will be reviewing your project on their own PC.
6. In case of non-compliance, the assignment will not be graded.
7. Your assignment will be graded on the attached rubrics (see end of document).

Deliverables

- Submit the following latest by **Week 15** (exact date will be communicated later):
 - Code of the application. You are required to bring the .cpp files for final assessment. Bring your laptop with custom installations.
 - Report in hard copy format organized as follows:
 - Text containing the following (organize in headings):
 1. Problem Description
 2. Distinguishing features of your project / list of object-oriented features implemented
 3. Flow of your project including class diagram. (To be submitted after Ramadan Break)
 4. Most challenging part for you while working on the project
 5. Any new thing learnt in C++ while working on the project
 6. Individual contributions of each member in the project
 7. Future expansions, if any.
 - At least 3 test case runs. Screen shots can also be attached.
 - While compiling the report, keep the following in mind:
 - Attach the provided rubric on top of your report, with names and roll numbers filled in.
 - No need for any other title page, the rubric sheet will serve as the title page.
 - Use font size 11 for regular text.
 - Use font size 14 for headings and 12 for subheadings
 - Font style should be Times New Roman.

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TERM PROJECT

Term Project Title:

LIBRARY AND RESOURCE MANAGEMENT SYSTEM

Description:

This system manages library resources and member activities by tracking resource availability, borrowing history, and fine calculation. It aims to improve efficiency and accuracy in library operations.

Minimum Required Features:

Your application should contain the following:

- A. Store and display details of library resources such as title, author, category (literature, science & technology, arts, education etc.), and availability status.
- B. User should be shown only those resources which are available for borrowing.
- C. Allow user to create an account with some basic information like username, password, first and last names, address, balance etc.
- D. Allow user to login to the respective account and borrow a resource based on available resources.
- E. User cannot borrow more than two resources at a time (means on the same day).
- F. Maintain borrowing history for each user with issue date, due date, resource issued, and any other relevant information.
- G. If a user fails to return a resource on time, a fine is deducted from his/her balance.
- H. User can search for resources based on different criteria.
- I. The system provides the customer with an interface to perform all operations described above, review borrowing history and update account balance.
- J. An administrator can print a report about all customers and their borrowing history including their current borrows.
- K. An administrator can also add new resources, update or remove resources from the system.
- L. An administrator can also print a report about all resources that are currently issued or overdue.

Design Hints:

1. Finalize the user interface for your application.
2. Choose data structures and classes to keep and organize the information when your application is running.
3. Decide on information that needs to be stored in files, for e.g. user accounts info, issue history etc.
4. Implement the application class by class.
5. Mention constraints or any other design scenario clearly.

Some Extension Ideas:

- Use GUI to improve interface for customer and administrator
- Make web interface for the application.
- Extra Classes to increase functionality of the system can be done as well.
- Support multiple resource types with different borrowing rules.
- Allow user to have different types of memberships based on their borrowing history.
- Include priority handling when issuing resources.
- Include a function where if a user donates a number of books, based on the quantity of the resources, their membership is upgraded.
- Include a function where if a user returns a resource before the end date but with damages, a damage fee is deducted from their balance.
- Include a function where a user can reserve a resource for later.

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Grading Rubric

TERM PROJECT

Student No.	Name	Roll No.
S1		
S2		
S3		
S4		

CRITERIA AND SCALES					Marks Obtained			
					S1	S2	S3	S4
Criterion 1: Does the application meet the desired specifications and produce the desired outputs? (CPA-1, CPA-3)								
1	2	3	4					
The application does not meet the desired specifications and is producing incorrect outputs.	The application partially meets the desired specifications and is producing incorrect or partially correct outputs.	The application meets the desired specifications but is producing incorrect or partially correct outputs.	The application meets all the desired specifications and is producing correct outputs.					
Criterion 2: How well is the code organization?								
1	2	3	4					
The code is poorly organized and very difficult to read.	The code is readable only to someone who knows what it is supposed to be doing.	Some part of the code is well organized, while some part is difficult to follow.	The code is well organized and very easy to follow.					
Criterion 3: How friendly is the application interface? (CPA-1, CPA-3)								
1	2	3	4					
The application interface is difficult to understand and use.	The application interface is easy to understand and but not that comfortable to use.	The application interface is very easy to understand and use.	The application interface is very interesting/ innovative and easy to understand and use.					
Criterion 4: How does the student performed individually and as a team member? (CPA-2, CPA-3)								
1	2	3	4					
The student did not work on the assigned task.	The student worked on the assigned task, and accomplished goals partially.	The student worked on the assigned task, and accomplished goals satisfactorily.	The student worked on the assigned task, and accomplished goals beyond expectations.					
Criterion 5: Does the report adhere to the given format and requirements?								
1	2	3	4					
The report does not contain the required information and is formatted poorly.	The report contains the required information only partially but is formatted well.	The report contains all the required information but is formatted poorly.	The report contains all the required information and completely adheres to the given format.					
Total Marks:								